

Fine-Tuning Transformer Model for Narrative Podcast Summarization

A Case Study on Summarizing “This American Life” Episodes with T5

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Natural Language Processing
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Style-Aware Podcast Summarization

- **Takes a full podcast transcript and generates a short summary in the narrative style of *This American Life***
 - “People on a mission to achieve their goals before their window of opportunity closes.”
 - “A show about people who are suddenly confronted with who they are.”
- **Combines abstractive summarization with style transfer to produce concise, engaging descriptions**
 - Audience Engagement
 - Content Discovery
 - Automation
- **Goes beyond generic summaries; captures tone and personality**
 - Not designed to be a CliffNotes-style summary

Motivation



Goals

- **Manual summaries are time consuming**
- **Existing summarizers lack personality**
- **Podcast transcripts are long and conversational**
- **Consistent style is valuable**

- **Train a model** to mimic the human written style of *TAL* summaries
- **Produce concise, compelling descriptions** (1-2 sentences)
- **Maintain narrative tone** while condensing content
- **Evaluate with both metrics and human judgement** to ensure quality

The *This American Life* Podcast Transcript Dataset

Source CSVs:

- **lines_clean.csv:** individual lines separated by speaker; merged lines to form full transcript per episode
- **episode_info_clean.csv:** episode number, title, and broadcast date
- **episodes_summaries.csv:** created via custom Python web scraper pulling human-written summaries from *This American Life* website

episodes_summaries_full.csv

column	description
episode_id	ordered chronologically 1 - 630
radio_date	broadcast date
title	episode title
url	link to episode
summary	human written short description
transcript	full episode transcript

		B	C	D	E	
1	Name Box	io date	title	url	summary	transcript
2		1995-11-17	New Beginnings	https://www.thisamericanlife.org/1/new-beginnings	Our very first broadcast.	Joe Franklin? I'm ready. It's Ira Glass here. Oh, you're the emcee on the show
3		1995-11-24	Small Scale Sin	https://www.thisamericanlife.org/2/small-scale-sin	Three teenage boys tell us about their forays into low-level credit card hacking.	OK, three boys, aged 13, 15, and 16. All three chose to appear with fake nam
4		1995-12-01	Poultry Slam 1995	https://www.thisamericanlife.org/3/poultry-slam-1995	Decrying the wonders of turkeys, chickens, and other fowl.	In Danielle's house, ever since she was a girl, when dinner comes, sometime
5		1995-12-08	Vacations	https://www.thisamericanlife.org/4/vacations	Vacations gone awry—or perhaps vacations that never should have happened.	The thing about Hawaii is that before you go all anybody will say to you is, it's
6		1995-12-15	Anger and Forgiveness	https://www.thisamericanlife.org/5/anger-and-forgiveness	Stories that reveal the societal "trend" toward anger and away from genuine forgiveness.	Hi. It's Ira Glass recording this in 2006. And the show that you're about to list
7		1995-12-22	Christmas	https://www.thisamericanlife.org/6/christmas	A Christmas radio play by David Sedaris.	From the WBEZ Chicago, it's Your Radio Playhouse. Hi. My name is Rever
8		1995-12-29	Quitting	https://www.thisamericanlife.org/7/quitting	People who quit everything in their lives that they hated.	What will make you change? For most of us, it is not self-help books, course
9		1996-01-03	New Year	https://www.thisamericanlife.org/8/new-year	Reflecting back on 1995.	What's the French electrical supply like over there? It's good actually. All the
10		1996-01-10	Julia Sweeney	https://www.thisamericanlife.org/9/julia-sweeney	Julia Sweeney and her brother found out they each had cancer six months apart.	OK, here's something that most of us would never, ever do. When Julia Swe
11		1996-01-17	Double Lives	https://www.thisamericanlife.org/10/double-lives	People living their lives, just like the rest of us. Plus an extra life.	There's what we wish for and there's what we get. For Susan Bergman, the st
12		1996-01-24	Enemies	https://www.thisamericanlife.org/11/enemies	A friendship between two young boys that was destroyed through the manipulative acts of one of them.	"I'll pour this pestilence into his ear. So will I make the net that will enmesh t
13		1996-01-31	Animals	https://www.thisamericanlife.org/12/animals	The animalness of animals, and the irreducible ways in which they are not human.	From PRI, Public Radio International. From PRI, Public Radio International. F
14		1996-02-07	Love	https://www.thisamericanlife.org/13/love	The dream of true love and the difficulties with achieving and maintaining that dream.	This is Ira Glass, host of Your Radio Playhouse, and starting this week, WBE
15		1996-02-21	Accidental Documentaries	https://www.thisamericanlife.org/14/accidental-documentaries	Stories made from old recordings found in attics and thrift stores.	From PRI, Public Radio International. From PRI, Public Radio International. F
16		1996-02-28	Dawn	https://www.thisamericanlife.org/15/dawn	Jack Hitt stumbles onto an epic story of the Old South and changing cultural norms in America.	From PRI, Public Radio International. From PRI, Public Radio International. F
17		1996-03-14	Economy	https://www.thisamericanlife.org/16/economy	Politics, the economy, and the big picture.	Well, in this moment of our presidential election process, let us consider the p
18		1996-03-21	Name Change / No Theme	https://www.thisamericanlife.org/17/name-change-no-theme	Our first show as This American Life.	From WBEZ Chicago, it's Ira Glass and His Radio Cowboys. [MUSIC - "BON
19		1996-03-28	Lies	https://www.thisamericanlife.org/18/lies	A girl who adored her father tries to figure out what to think after he takes some of her college money aw	You know, if you think about it, the idea of April Fools' Day is just absurd, a d
20		1996-04-12	Rich Guys	https://www.thisamericanlife.org/19/rich-guys	Men who had comfortable, decent lives, yet decided to do something wild and eccentric.	If you think about it, it's a strange thing for a very rich man to do to run for pr
21		1996-04-19	From A Distance	https://www.thisamericanlife.org/20/from-a-distance	Stories of admiring someone from afar, and trying to get closer to them.	From PRI, Public Radio International. From PRI, Public Radio International. F
22		1996-04-26	Factions	https://www.thisamericanlife.org/21/factions	Friends get together to start a business, start a church, do political action together. And after a while, th	It is an old, old story. A group of friends gets together, decides to start a busi
23		1996-05-03	Adult Children	https://www.thisamericanlife.org/22/adult-children	Stories of the difficult relationships between parents and their grown children.	When I picked up the phone, it was my mom. And it had been about a month
24		1996-05-10	Drama Bug	https://www.thisamericanlife.org/23/drama-bug	Stories of the people who fall for a life in the theater.	Sure, back when they were freshmen, Kim and Tiffany and Laura were starst
25		1996-05-24	Teenaged Girls	https://www.thisamericanlife.org/24/teenaged-girls	Girls who have to figure out how they're going to act when the ground rules are constantly shifting.	All right, now this moment from the life of a teenage girl in these United State
26		1996-06-07	Basketball	https://www.thisamericanlife.org/25/basketball	The Chicago Bulls and their grip on Chicagoans' hearts and lives during the NBA playoffs.	The story goes like this. Recently I heard about this couple. He's in his 40s, s
27		1996-06-14	Father's Day '96	https://www.thisamericanlife.org/26/fathers-day-96	Ira's own father, Barry Glass, co-hosts this special Father's Day show.	OK, so Dad, so you have the script. I have the script. From WBEZ Chicago it
28		1996-06-21	The Cruelty of Children	https://www.thisamericanlife.org/27/the-cruelty-of-children	A story of youthful backstabbing by David Sedaris.	From PRI, Public Radio International. From PRI, Public Radio International. F
29		1996-07-12	Detectives	https://www.thisamericanlife.org/28/detectives	David Sedaris recalls the days when his mother and sister played armchair detective.	From WBEZ Chicago and Public Radio International, it's This American Life. I
30		1996-07-19	Bob Dole	https://www.thisamericanlife.org/29/bob-dole	Unusual perspectives on the presumptive Republican nominee.	From WBEZ Chicago, it's This American Life. I'm Ira Glass. Happy birthday to
31		1996-07-26	Obsession	https://www.thisamericanlife.org/30/obsession	What happens when a little idea starts to control you.	So Paul, let's begin this week's show this way. I want you to explain to me t
32		1996-08-09	When You Talk About Music	https://www.thisamericanlife.org/31/when-you-talk-about-music	Stories of people whose lives are transformed by music.	From WBEZ Chicago, it's This American Life. I'm Ira Glass. And each week
33		1996-08-16	Republican Convention	https://www.thisamericanlife.org/32/republican-convention	Unusual stories about day-to-day life at the Republican Convention.	The trick to getting a politician's autograph, John Rossi tells me, is to try to c
34		1996-08-23	A Night at the Wiener Circle	https://www.thisamericanlife.org/33/a-night-at-the-wiener-circle	Ira Glass spends a night at one of the most boisterous hot dog stands in the nation: Chicago's Wiener C	Hi. It's Ira, recording this in 2006. This episode of our show, Episode 33, is or
35		1996-08-30	Democratic Convention	https://www.thisamericanlife.org/34/democratic-convention	Unusual stories from the 1996 Democratic Convention in Chicago.	Sure, the Democrats are doing great in the polls this week, but to consider wh
36		1996-09-06	Fall Clearance Sale	https://www.thisamericanlife.org/35/fall-clearance-sale	Stories that haven't fit into any of our regular "theme" shows.	Okay, so I'm standing here on Navy Pier in Chicago. And could I just ask you
37		1996-09-20	Letters	https://www.thisamericanlife.org/36/letters	We took out advertisements in Chicago inviting people to share letters they've received, sent, or found.	From PRI, Public Radio International. From PRI, Public Radio International. F
38		1996-09-27	The Job That Takes Over Your Life	https://www.thisamericanlife.org/37/the-job-that-takes-over-your-life	Scott Carrier got a job interviewing schizophrenics. After awhile, he began to wonder if he was a schizop	Maupassant didn't care much for the Eiffel Tower. And he didn't care for the fi
39		1996-10-11	Simulated Worlds	https://www.thisamericanlife.org/38/simulated-worlds	Civil War reenactments, wax museums, simulated coal mines, fake ethnic restaurants, an ersatz Medie	From PRI, Public Radio International. From PRI, Public Radio International. F
40		1996-10-25	Halloween	https://www.thisamericanlife.org/39/halloween	Stories of things that are supposed to be scary, but aren't.	Every Monday, John, and Thax, and Ericka get together at John's apartment
41		1996-11-01	Lessons	https://www.thisamericanlife.org/40/lessons	Spalding Gray on finding the zen of skiing.	From PRI, Public Radio International. From PRI, Public Radio International. F
42		1996-11-08	Politics	https://www.thisamericanlife.org/41/politics	Stories of the election you can't hear anywhere else.	So what was it like after election day this week for the couple who live in the
43		1996-11-15	Get Over It!	https://www.thisamericanlife.org/42/get-over-it	Trying to forget the past and move on.	From PRI, Public Radio International. From PRI, Public Radio International. F
44		1996-11-22	Faustian Bargains	https://www.thisamericanlife.org/43/faustian-bargains	People who made a deal with the devil.	So LuAnne Johnson was a teacher, and she wrote a book about her job, and
45		1996-11-29	Poultry Slam 1996	https://www.thisamericanlife.org/44/poultry-slam-1996	Stories about turkeys, chickens, ducks, and fowl of all kinds.	We're in Danielle's house. Ever since she was a girl, when holidays dinners co
46		1996-12-06	Media Fringe	https://www.thisamericanlife.org/45/media-fringe	People struggling at the fringes of our nation's infotainment industry.	From PRI, Public Radio International. So Sandra thought she knew how to be c

Project Approach

Initial Approach

- **Goal:** Generate summaries directly from full transcripts using pre-trained *pszemraj/led-large-book-summary* model
- **Result:** Very long (3,000+ word) summaries

Pivot

- Matched and trained each transcript with its official short summary using *t5-base*
- Experimented with chunking inputs to ingest full transcript
- Nature of material was accurate even with 1,024 token limit

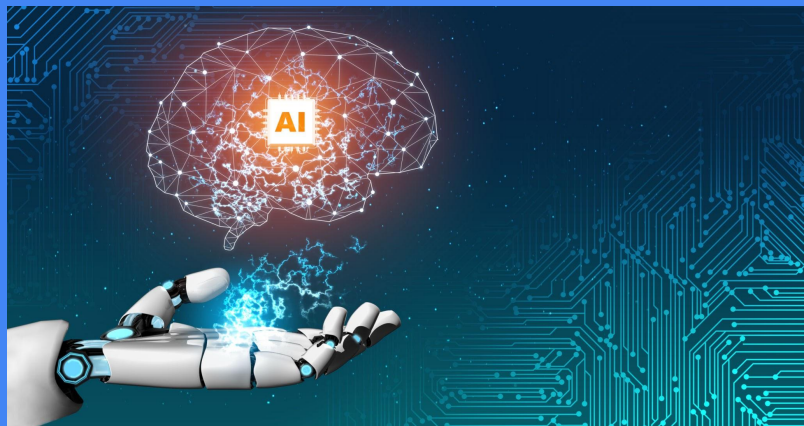
Implementation

- Preprocessed the transcripts and summaries
- Created the train/test split
- Fine-tuned the transformer model
- Evaluated results with ROUGE and BERTScores

- **Hugging Face Transformers:** fine-tuning and inference for T5-base
- **Hugging Face Datasets:** dataset creation, train/test split
- **Pandas:** CSV loading/merging/cleaning
- **tqdm:** progress bars
- **T5-Base:** abstractive summarization model
- **BERTScore:** semantic similarity metric
- **ROUGE:** n-gram overlap
- **Googel Colab + A100/L4:** training/inference in the cloud, utilization of GPUs
- **NLTK:** tokenization and preprocessing



Tools & Techniques

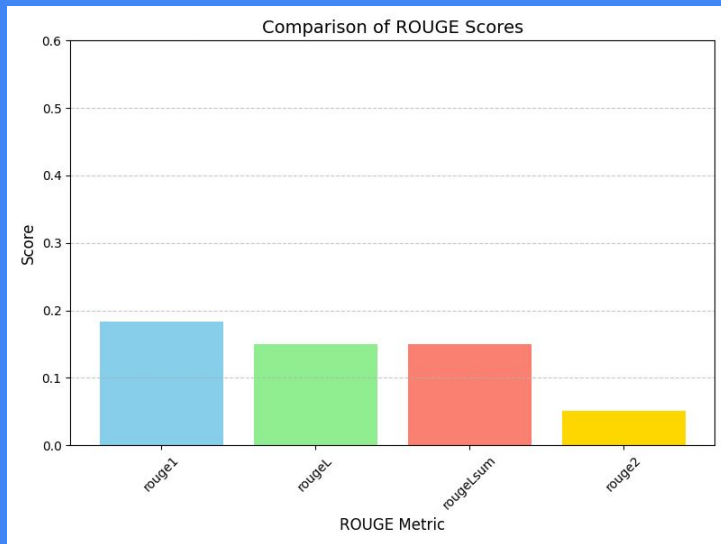


What I Tried

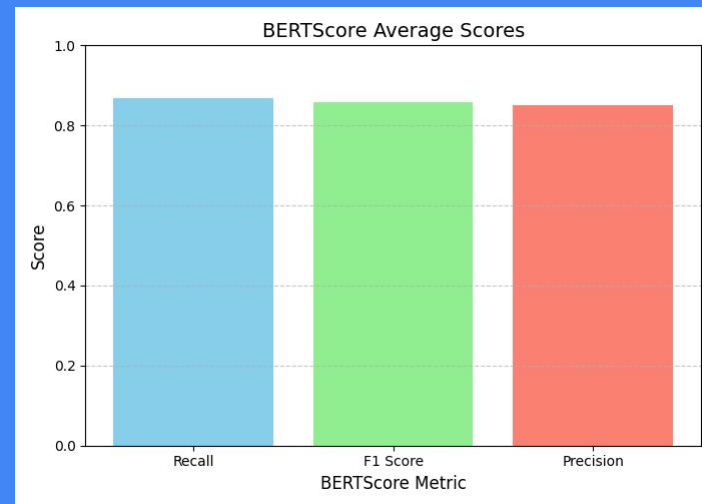
Outcome

Notes / Lessons Learned

Summarizing full episode transcripts	Very long summaries (3,000+ words)	Too long for most readers; not engaging
Gradually shortening outputs	More concise, easier to read	Led to idea of mimicking <i>TAL</i> human summaries
Mimicking <i>TAL</i> human-written summaries	Best reader engagement style	Matches tone and brevity of official summaries; Needs more tweaking/training to improve quality
Testing other models (t5-large, t5-small)	t5-base performed best	t5-large ran into errors and poor summaries
Chunking transcripts to bypass 1,024 token limit	Anecdotally worse results	Realized intro usually contains full summary
Using only first 1,024 tokens	Faster training	Retains most key summary info since <i>TAL</i> intros are comprehensive



- ROUGE-1 has a low level of single word overlap
- ROUGE-2 also has a low level of two word overlap
- Low similarity in longest common subsequence
- Low results here are expected based on model goals



- Recall shows that a substantial portion of the key information is semantically captured by the generated summaries
- F1 represents a strong balance between Precision & Recall
- Precision indicates that a high percentage of the words generated are semantically similar to the reference summaries

The Good

Original Summary	Generated Summary
Discovering just how much time members of Congress spend raising money.	The world of lobbyists and political fundraising. A world that we all hear about all the time, but most of us never have seen up close.
Civil War reenactments, wax museums, simulated coal mines, fake ethnic restaurants, an ersatz Medieval castle, and other re-created worlds that thrive all across America.	In this week's program, we take a whack at the theme of reenacting the Civil War: The American imagination demands the absolute fake.
Stories of things that are supposed to be scary, but aren't.	Stories that are supposed to be scary, but aren't. Act one, Dark Shadows. Act two, Vampire Girl.
Spalding Gray on finding the zen of skiing.	In this week's episode of This American Life, we bring you a story from Spalding Gray, who is sometimes called the premier teller of monologues in America.

The Bad

Original Summary	Generated Summary
The dream of true love and the difficulties with achieving and maintaining that dream.	Ira Glass, host of Your Radio Playhouse, rebroadcasts her program on Friday nights at 8 o'clock.
Men who had comfortable, decent lives, yet decided to do something wild and eccentric.	This American Life is a weekly radio show documenting life in the United States through whatever storytelling means we believe will amuse you and yourself.
Stories of people who try to convince you that the devil is there, whispering in your ear.	What happens when a bicycle flips on its side at a very high speed, throwing the bicyclist off and bruising him.
Sarah Koenig's mother lives by a set of rules about conversation. We tried to prove her wrong.	We're all boring on occasion. But we're not always forgiven. This American Life is about my mother, Maria Matthiessen.

Closing Thoughts

- **Balancing Ideas & Execution**
- **Strategic Use of Compute Resources**
- **Navigating Project Scope Creep**
- **Iterative Nature of Improvement**